Research assessment 1.
Foundations of Data Analytics and Data Science.
Boitumelo Themba
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Section A: Data Fundamentals

1. Database - A database is an organised collection of data stored electronically so it can be easily accessed, managed and updated.
Importance: It helps businesses store, retrieve and manage large amounts of information effectively.
Examples:
* Banking systems (store customer accounts and transactions)
* E-commerce websites (store products, orders and customer data)

2. Data Warehouse - A data warehouse is a system used to store large amounts of historical data for analysis and reporting.
The difference is that a database is for day-to-day operations and a data warehouse is for analysis and reporting.
Why a company would use it is to help them make better business decisions using historical data.

3. Data Lake - A data lake stores raw data (structured, semi-structured and unstructured) in its original form
The difference is that a data warehouse is structured and it uses processed data whereas a data lake uses raw and flexible data.
It is used when businesses need to store massive amounts of diverse data.

4. Data Mart - A smaller set / subset of a data warehouse focused on specific departments (e.g., sales).
The users of this type of data are department teams: marketing and finance.

5. Data Types:
* Structured: Organised (tables, rows) e.g Excel spreadsheet.
* Semi-structured: It has some structure e.g JSON/XML
* Unstructured: No structure e.g videos, emails

6. Data Modeling - Data modeling is a process of designing how data is stored and organised.
The importance of this is to ensure efficiency and accuracy
The types of data modeling are:
* Conceptual model
* Logical model.

7. Data mining - Data mining is defined as finding patterns in large datasets. For example:
* Netfix recommending movies
* Banks detecting fraud.

8 Database Management System (DBMS) - This is a software used to manage databases. It helps users create, manage and interact with databases. Example:
* MySQL - is used for websites and simple applications (e.g blogs)
* PostgreSQL - advanced database for complex queries and large systems.
* Oracle Database - Used by large organisations for high-performance systems.
* MongoDB - No SQL database used for flexible unstructured data (e.g apps with changing data)

9. ETL (Extract, Transform, Load) - this is a process used to prepare data for analysis.
* Extract: Collect data from different sources
* Transform: Clean and format the data
* Load: Store it in a database or warehouse

Importance of this is that it ensures that the data is acurate and ready for decision making.

10. Data Storage Formats:
* CSV - Simple text format for tables
* JSON - Structured format used in APIs
* Parquet - Efficient format for big data and analytics.
* Avro - Used for data streaming and large systems.

Section B

11. Artificial Intelligence (AI) - AI is the ability of machines to perform tasks that require human intelligence.
* Narrow AI: Designed forone task (e.g Siri)
* General AI: Can perform any task like a human (not yet fully developed)

12. Machine Learning (ML) - ML allows computers to learn from data instead of being programmed.

Types of machine learning:
* Supervised learning: Uses labeled data
* Unsupervised learning: Finds patterns in data
* Reinforcement learning: Learns through rewards and punishments.

13. Deep Learning - A type of machine learning that uses neural networks to learn complex patterns.

* Network learning - systems inspired by the human brain that process data in layers.

14. Natural Language Processing (NLP) - Technology that helps computers understand human language. e.g Chatgpt, Google translate and voice assistants (Alexa, Siri)

15. Large Language Models (LLMs) - LLMs are AI models that are trained on large text data. e.g Chatgpt - generates texts and answers questions. Claude is another example because it assists with writing and reasoning.

16. Generative AI - This is AI that generates or creates new content like text, images or audio.

* Tools:
- Chat gpt → text
- DALL-E → images
- Midjourney → artwork

17. Training dataset - A dataset that is used to train an AI system. The importance of this is:

* Good data - accurate results
* Bad data - incorrect or biased results

18. Computer Vision - This is technology that allows machines to interpret images.

* Examples:
- Facial recognition
- Self-driving cars

Section C

19. API (Application Programming Interface) - An API allows different systems to communicate. The importance of an API is that it helps apps share data easily for example:
* Like a waiter connecting customers and the kitchen.

20. API Key - A unique code used to access an API. The importance of this is that it keeps access secure and makes sure it is private.
The risk is that if it is exposed others can misuse the API

21. REST API - This is a type of API that uses HTTP methods. The REST API (Representational State Transfer API) is a type of web service that allows different systems or applications to communicate with each other over the internet using the standard HTTP method.
Main HTTP methods in the REST APIs:
- Get: used to retrieve data from a server e.g getting a list from a database of users.
- Post: used to create new data e.g adding new user to the system
- PUT: used to update existing data e.g changing data like a user's email address
- DELETE: used to remove data e.g deleting a user account

22. JavaScript Object Notation (JSON) - is a simple format used to store data between systems especially in web APIs and applications. Common uses are: easy for humans to read and write, easy for computers to process and works well with APIs and web services. e.g JSON is a lightweight format used to send and store data in a structured way.

23 Prompt Engineering - this is the process of writing clear and effective instructions for AI tools like Chatgpt. This is important because it helps get accurate and useful answers and improves the quality of AI responses. e.g Poor promp: "Tell me about business" and Improved prompt: "Explain small business trends in 200 words whith examples"

24 Map function: A map function applies a specific operation to each item in a list. What the map function does is that it multiplies each number by 2 and the output will therefore be: [2, 4, 6, 8, ...] For example:

1. Number = [1, 2, 3, 4]
2. Results = list(map(lambda x: x*2, numbers))

25 Cloud computing - cloud computing is the delivery of computing services (like storage and software) over the internet.

Three main types

- IaaS (Infrustructure as a Service) - provider virtual servers and storage e.g AWS EC2
- PaaS (Platform as a Service) provide platform to build an application e.g Google App Engine.
- SaaS (Software as a Service) - provides ready to use software - e.g Microsoft 365

26 Version Control System - this tracks and manages changes to code over time. Git is a popular version control tool used by developers and data teams. Commit - Saving changes to a project. Branch - creating a separate version of the code

27 Jupyter Notebook - it is an interactive tool used to write and run code, mainly for data analysis.
Common language is ~~ph~~ Python.
This Jupyter Notebook is popular because it combines code, text and visuals in one place. It is also easy for analysis and experimentation.

28 Python - this is a popular programming language used in data analytics and AI. It is popular because it is easy to learn, powerful for data analysis and large community analysis.
Common libraries:
* Pandas - data analysis
* NumPy - mathematical operations
* Matplotlib - data visualisation (charts)
* Scikit-learn - machine learning

29 Data Privacy - this refers to the protection of personal information from being accessed, misused or shared without permission. This is important because it protects sensitive information e.g ID, banking details. It also prevents identity theft and fraud and it also helps build trust between businesses and users.
* GDPR - General Data Protection Regulation is a law that protects personal data and privacy. It applies in European Union
* It gives people control over their personal data.

30 Data Ethics - it is about using data responsibly and fairly ~~eg~~ selling personal data without concent, using biased AI hiring, creating fake accounts. All these are examples of unethical data use. This is a problem because it violates trust, leads to unfair treatment and can harm individuals or groups.

31. Data Governance - this is the managment of data to ensure it is accurate, secure and properly used.
Key components:
1. Data Quality: data must be ~~used~~ accurate and reliable.
2. Data Security: Protect data from unauthorised access
3. Compliance: follow laws and regulations

32. Bias AI - this in AI occurs when an AI system produces unfair or inaccurate results due to biased data. For example a hiring system that prefers certain genders, ages or races ~~baised~~ biased traing data.
Consequences:
1. Discrimination
2. Unfair decisions
3. Loss of trust in AI systems.

33. Differences between Data Roles

	Role	Focus	Skills	Tools
1.	Data Analyst	Analyse data	Excel, SQL	Power BI
2	Data Scientist	Build models	Python, ML	Tensor Flow
3	Data Engineer	Build data systems	Big data tools	Hadoop
4.	AI Engineer	Develop AI systems	Deep learning	Py Torch

34. Business Intelligence (BI) - this uses data to help businesses make better decisions. Popular tools: Power BI → dashboard and reports. Tableau → data visualisation. How it helps is that it identifies trends, improves decision making and tracks business performance.

Section E

35 I found Generative AI the most interesting because it ~~came~~ can create content like text, images and videos. This shows how advanced technology has become and how it can be used in many industries. This shows how AI is rapidly changing how we create and use information in everyday life.

36 I found data modeling difficult at first because it involves planning how data is structured. I used online tutorials, videos and practice examples to understand it better. With practice and research, complex data can become easier to understand over time.

37 Data and AI have and will have an impact on my future career and daily life. For example:
* I can use AI to complete and finalise data faster.
* Automation can reduce manual work.
* Better decision-making using data insights.

Overall, data and AI will play a key role in improving overall efficiency and decision-making in the future.

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